

2. Report on results of capacitor demonstration and village response

2.1 Structured demonstration of capacitors in two districts

Capacitor demonstration was done in 10 villages of two districts beed and Wardha District. As mentioned in the Phase III report, the capacitor demonstration was done systematically. It included the following activities:

- Identification of village through secondary Data
- DTs selection through village overview meeting
- Training workshop of concerned stakeholders (AE/JE, Lineman, AA, Farmers etc)
- Introductory meetings in villages to introduce capacitors
- Capacitor demonstration drive on overloaded DTs (Measurement on DTs)
- Installation of capacitors
- Spreading of awareness and communication material (Film, Banner, Pamphlets etc)
- Survey to estimate the effect of capacitor demonstration
- Village meeting post rabi to highlight the impact of capacitor

2.1.1 Capacitor survey to estimate the effect of capacitor demonstration

Farmers were surveyed at the end of Rabi to estimate the capacitors' effect. The surveys were conducted to know

- Changes in pump breakdowns, number of tripping etc.
- Change in the number of hours of uninterrupted supply of electricity
- The DT failure rate in peak season
- Farmers' satisfaction level with capacitor usage
- Benefits of capacitor usage

For the survey, a detailed questionnaire was designed. The questionnaire was added in the present report as Annexure A. Kobo toolbox was used to conduct a survey. A few farmers on each DT were surveyed and their response was noted.

Through the survey, the effect of the capacitor on the irrigation pump before and after the installation of the capacitor was noted. Apart from that farmers' satisfaction and awareness about capacitors were also noted. In total 98 farmers were surveyed; out of which 41 Farmers were from Wardha district and 57 were from Beed district. Out of the selected villages, only Vashist Kedar DT farmers were not surveyed in detail because it was observed that due to the unavailability of water farmers didn't install capacitors on their pumps. Table 2.1 gives the details of the conducted survey on selected DTs:

Table 2.1. Details of selected DTs for Beed District

Sr. No.	Village	Name of the DT	kVA rating	No. of Pumps	Present capacitor Users	Number of farmers surveyed
1	Bagpimpalgaon	Canal DT	100 kVA	18	15	10
		Umapur fata DT	63 kVA	14	14	7
2	Rui	Baman mala DT	63 kVA	21	14	5
3	Tigaon	Bag DT	100 kVA	31	31	13
4	Shivni	Kakarwadi DT	63 kVA	22	19	13
5	Pimpalwandi	Babu Shinde DT	100 kVA	13	13	9
6	Ekruka	Vashisht Kedar DT	100 kVA	31	25	-

Table 2.2. Details of selected DTs for Wardha District

Sr. No.	Village	Name of the DT	kVA rating	No. of Pumps	Present capacitor Users	Number of farmers surveyed
1	Kanholi	Singhan DT	100 kVA	12	11	6
2	Rohana	Raut DT	100 kVA	19	17	12
3	Nandpur	Dondal DT	100 kVA	18	16	11
4	Junona	Giri DT	100 kVA	24	22	12

2.2 Results of Capacitor Demonstration

Based on data collected through the capacitor survey the impact of structured demonstration of the capacitor was studied. In total 98 farmers were surveyed; out of which 41 Farmers were from Wardha district and 57 were from Beed district. Mainly quantitative analysis was done to understand the status of capacitor usage, the effect of capacitor usage and satisfaction level in terms of the overall performance of the pump after capacitor installation among farmers.

2.2.1 Status of Capacitor Usage

The previous studies of IIT Bombay highlighted that farmers remove installed capacitors from their pumps if they do not have a proper understanding of capacitors. So after a year-long demonstration of the value of capacitors, it is necessary to check the present status of capacitor usage.

The present status of capacitor usage on irrigation pumps collected through the survey is given in Table 2.3. The survey results show that most of the farmers installed capacitors on their Pumps. Few farmers have not installed the capacitors on their pumps.

Table 2.3 Status of capacitor usage among surveyed farmers

	Number of Surveys	Capacitor installed on Pump		Capacitors are still in use		Reason for not installing a capacitor		
		Yes	No	Yes	No	Didn't have Capacitor	Inert attitude	The pump is not in use
Dondal DT, Nandpur	11	9	2	9	0	2	0	0
Giri DT, Junona	12	7	5	6	1	0	5	0
Raut DT, Rohana	12	12	0	12	0	0	0	0
Sighan DT, Kanholi	6	4	2	3	1	0	2	0
Babu Shinde DT, Pimpalwandi	9	7	2	7	0	2	0	0
Bagh DT, Tigaon	13	9	4	4	5	4	0	0
Baman mala DT, Rui	5	1	4	0	1	2	0	2
Canal DT, Bagpimpalgaon	10	8	2	5	3	2	0	0
Kakarwadi DT, Shivni	13	11	2	4	7	1	0	1
Umapur Fata, Bagpimpalgaon	7	4	3	2	2	1	2	0

The different reasons were identified through a survey related to the non-installation of capacitors are :

1. Few farmers didn't take capacitor
2. The pump not in use due to the non-availability of water (Pumps on Dam/Canal/Streams)
3. Inert attitude (Not concerned about capacitor benefits or fewer problems compared to others)

There are also cases where farmers removed installed capacitors from their pumps. Such cases were observed mostly on DTs where seasonal connections are there. On such DTs, pumps are used on water sources like streams, canals or rivers where water availability is also seasonal. So farmers remove the pump once the water source dries out. In some villages, Farmers tend to remove capacitors after Rabi irrigation is completed. It is because of fear of robbery among farmers themselves.

2.2.2 Results of a survey conducted to estimate the effect of capacitor

To quantify the effect of capacitor usage, a few parameters were identified based on previous experience and interactions with farmers. This effect on various parameters is tabulated in Table 2.4 for the Beed district and Table 2.5 in the Wardha district, and explained as follows:

Table 2.4: Results of a survey conducted on capacitor usage at Beed

	Number of Surveys		Hours of uninterrupted supply		Number of times pump tripping		Number of hours that pump ran		Number of times your DT burnt	
			Before	After	Before	After	Before	After	Before	After
Babu Shinde DT, Pimpalwandi	9	Avg	2.57	5.29	3.14	1.14	5.57	7.00	3.29	0.00
		Std Dev	1.27	0.95	1.07	2.19	0.53	0.00	0.49	0.00
Bagh DT, Tigaon	13	Avg	1.30	3.10	6.20	1.40	4.60	6.75	1.20	0.00
		Std Dev	0.57	1.02	2.28	1.14	0.55	0.65	0.45	0.00
Baman mala DT, Rui	5	Avg	1.00	1.00	5.00	5.00	4.00	4.00	1.00	0.00
		Std Dev	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Canal DT, Bagpimpalgaon	10	Avg	0.88	1.96	6.88	3.38	2.94	4.56	1.50	0.13
		Std Dev	0.44	0.84	2.30	2.07	0.62	0.94	0.76	0.35
Kakarwadi DT, Shivni	13	Avg	1.45	2.55	2.91	0.64	3.55	4.00	2.36	0.00
		Std Dev	1.21	1.21	1.22	1.03	1.04	1.48	0.50	0.00
Umapur Fata DT, Bagpimpalgaon	7	Avg	1.75	2.50	4.75	2.50	4.75	5.75	0.50	0.00
		Std Dev	0.50	1.00	0.96	1.29	0.96	1.50	0.58	0.00

Table 2.5: Results of a survey conducted on capacitor usage at Wardha

	Number of Surveys		Hours of uninterrupted supply		Number of times pump tripping		Number of hours Pump ran		Number of times your DT burnt	
			Before	After	Before	After	Before	After	Before	After
Dondal DT, Nandpur	11	Avg	11.11	12.44	2.78	0.22	12.00	13.22	2.00	0.00
		Std Dev	1.17	1.01	0.44	0.44	0.87	0.97	0.00	0.00
Giri DT, Junona	12	Avg	1.57	3.57	3.86	1.00	5.43	6.57	0.71	0.00
		Std Dev	0.45	1.40	0.69	0.58	1.62	1.51	0.49	0.00
Raut DT, Rohana	12	Avg	5.67	6.75	2.83	0.25	5.83	6.50	2.17	0.00
		Std Dev	0.98	0.97	1.03	0.45	0.39	0.52	0.39	0.00
Sighan DT, Kanholi	6	Avg	3.50	4.75	2.50	1.25	7.00	7.50	0.25	0.00
		Std Dev	0.58	0.96	0.58	0.50	1.15	1.00	0.50	0.00

a) Effect of capacitor usage on hours of uninterrupted supply

It can be observed from Tables 2.4 and 2.5, that the average hours of uninterrupted supply has increased substantially. However, the increase in standard deviation for most of the DTs shows, that the increase in average uninterrupted supply hours is not the same for all the pumps and this happens largely because of different pump ratings, location of pump connection on the grid and so on.

b) Effect of capacitor usage on the number of times pump trippings

It is observed that the decrease in an average number of pump trippings, however again the standard deviation for a few DTs shows, that the decrease in average numbers of trippings is not the same for all the pumps and this happens largely because of different pump ratings, location of pump connection on the grid and so on.

c) Effect of capacitor usage on the average number of hours pumps run per day

Most of the selected DTs have 6-8 hours of electricity supply per day except Nandpur village. It can be observed from Tables 2.4 and 2.5, that the average hours pump runs per day of electricity supply has been increased substantially. However, the increase in standard deviation for most of the DTs shows that the increase in average pump run hours is not the same for all the pumps and this happens largely because of different pump ratings, location of pump connection on the grid and so on.

d) Effect of capacitor usage on the number of times DT burnout

It is observed by farmers and IITB field staff that, this year after the installation of capacitors most of the DTs haven't recorded any such DT burnout incidences. Here, claiming directly that DT burnout incidences have been solely because of the usage of capacitors is not appropriate. Out of the different influencing parameters of DT burnout, low voltage and high current drawl from DT due to capacitor usage might have little impact.

2.2.3 Satisfaction level among farmers

During the survey, farmers were asked about their satisfaction level with the performance of the capacitor. Their responses were categorized into three levels: Good, Satisfactory, and Poor. The majority expressed satisfaction, reporting reduced electricity-related issues like low voltage, light tripping, fuse burning etc. post-capacitor installation. Some noted no observable change, attributing it to a low rainfall year and minimal load on their Distribution Transformers (DTs). Very few farmers provided negative feedback regarding the capacitor installation. The following figure shows satisfaction levels among farmers due to the usage of capacitors.

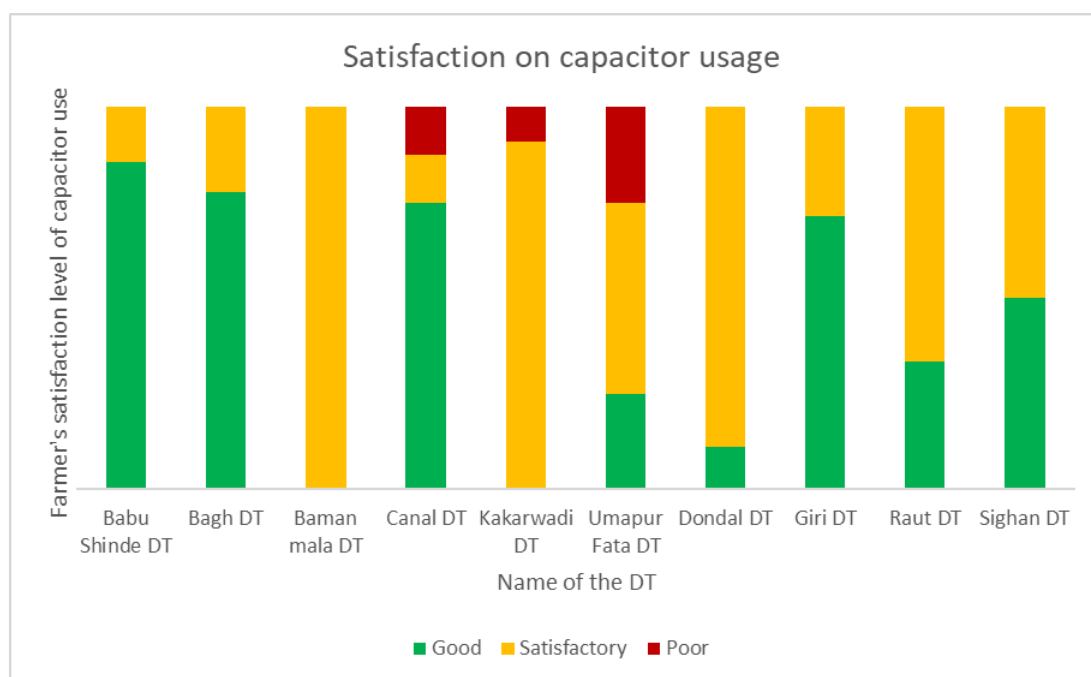


Figure 2.1 Satisfaction levels among farmers due to the usage of capacitor

2.3 Village meetings to present results of capacitor demonstration

2.3.1 Village Meetings in Beed and Wardha District

Village meetings were conducted in selected villages to demonstrate the effect of capacitors. Farmers on selected DTs were surveyed to note down pump breakdowns and present the status of capacitor usage. The results of the surveys were presented in village meetings. The meeting was not conducted in the two villages of Beed. In Rui village, the meeting was not conducted due to the political environment of the Loksabha elections while In Ekruka village it was mainly due to capacitors were not installed on pumps. Table 2.6 gives a timetable of meetings conducted in the Beed and Wardha Districts. The summary of village meetings will be given in the following subsections.



Figure 2.1 IITB team member presenting results of capacitor demonstration in Tigon village, Beed



Figure 2.2 IITB team member presenting results of capacitor demonstration in Rohana village, Wardha

Table 2.6 Details of village meetings to present results of capacitor demonstration

Sr. No	District	Village	Date	Attendance
1	Wardha	Rohana	11 March 2024	Farmers, Village representatives, IIT energy team, linemen
2	Wardha	Kanholi	12 March 2024	Farmers, Village representatives, IIT energy team, linemen
3	Wardha	Nandpur	13 March 2024	Farmers, Village representatives, IIT energy team, linemen

4	Wardha	Junona	7 March 2024	Farmers, Village representatives, IIT energy team, linemen
5	Beed	Pimpalwandi	24 April 2024	Farmers, Village representatives, IIT energy team, linemen
6	Beed	Shivni	19 April 2024	Farmers, Village representatives, IIT energy team, linemen
7	Beed	Bagpimpalgaon	24 April 2024	Farmers, Village representatives, the IIT water team, linemen
8	Beed	Tigaon	22 April 2024	Farmers, Village representatives, IIT energy team, linemen

2.3.2 Meeting Response

After the rabi season, Village meetings to highlight the effect of capacitors were conducted for selected DTs. This is the fourth village meeting for the capacitor demonstration. Earlier village meetings were - village overview meetings, introductory meetings, and meetings for conducting measurements.

As the general election for Lok Sabha is going on, it is recommended not to conduct official meetings in the villages. So government officials from agriculture and the MSSEDCL department were not involved in the village meeting to present the result of the capacitor demonstration. Farmers were mobilised with the help of village representatives and the IIT Bombay energy team only. The presence of farmers for the meeting was moderated compared to Rabi Hagram Baithaks. It is also because most of the farmers are now aware of capacitor usage.

2.3.2 Summary of Village Meetings

The points discussed in village meetings are as follows

- Overview of the process followed for structure demonstration of capacitors
- Result of capacitor demonstration - Status, effect of capacitor etc.
- Awareness building and way ahead

a. Presentation of results of capacitor demonstration

The results of the capacitor demonstration for a village are presented in the village meeting. Capacitor usage in the village was presented during the meeting. In villages having less capacitor usage, farmers were asked about the reasons and during the village meetings, the effect of capacitors like the number of pump tripping, the number of hours of uninterrupted supply of electricity, and the average number of hours of pump ran in the peak period for a particular village was presented.

b. Discussion on the effect of capacitor demonstration

During the village meetings, the effect of capacitors like the number of pump tripping, the number of hours of uninterrupted supply of electricity, and the average number of hours of pump ran in the peak period for a particular village was presented. Farmers were asked to validate the results of their village. The farmers on all DTs experienced results similar to the outcome of the survey. The level of farmer satisfaction was discussed so that other farmers could understand the benefits of capacitor usage.

It was highlighted in the meeting that a few farmers were still not using capacitors. The names of farmers who have not installed capacitors due to their inert attitude were pointed out during the discussion so that they could install capacitors. It was emphasised that capacitor usage is a community-level intervention so the involvement of all the farmers is necessary to get good results.

Many farmers from the village asked about the capacitor details for installing pumps. It was mainly because selected DTs have not failed in this Rabi season and most of the farmers are satisfied with the performance of the capacitor. Details like the capacitor rating for a particular pump HP and the installation process were explained to the farmer group who have asked for it.

c. Awareness creation

Even though the primary purpose of the village meeting was to present the result of the capacitor demonstration and understand farmers' response; capacitor awareness was also created. It was done purposefully because many farmers have not installed capacitors even though they have used pumps in Rabi season. In village meetings, those who benefited from capacitor usage were asked to share their experience.

d. Experience of energy team in conducting village meeting

Village meetings at the end of Rabi help to affirm the importance of capacitor usage in the minds of farmers. The IIT Bombay energy team has observed that the importance of capacitors reached across different sections of villages. Many farmers on other DTs acknowledged the potential benefits of capacitor usage and expressed readiness to utilize them, especially considering future improvements in pump usage when water scarcity concerns are reduced.